

WINDOWS 11 IT ADMINISTRATION & TROUBLESHOOTING LAB

Hands-on practice report | Personal lab project

Project Overview

This project documents hands-on practice with common Windows administration and troubleshooting tools inside a virtual machine. Activities included reviewing local users and groups, inspecting firewall rules, testing network connectivity, reviewing Windows event logs and services, and examining system performance and disk information.

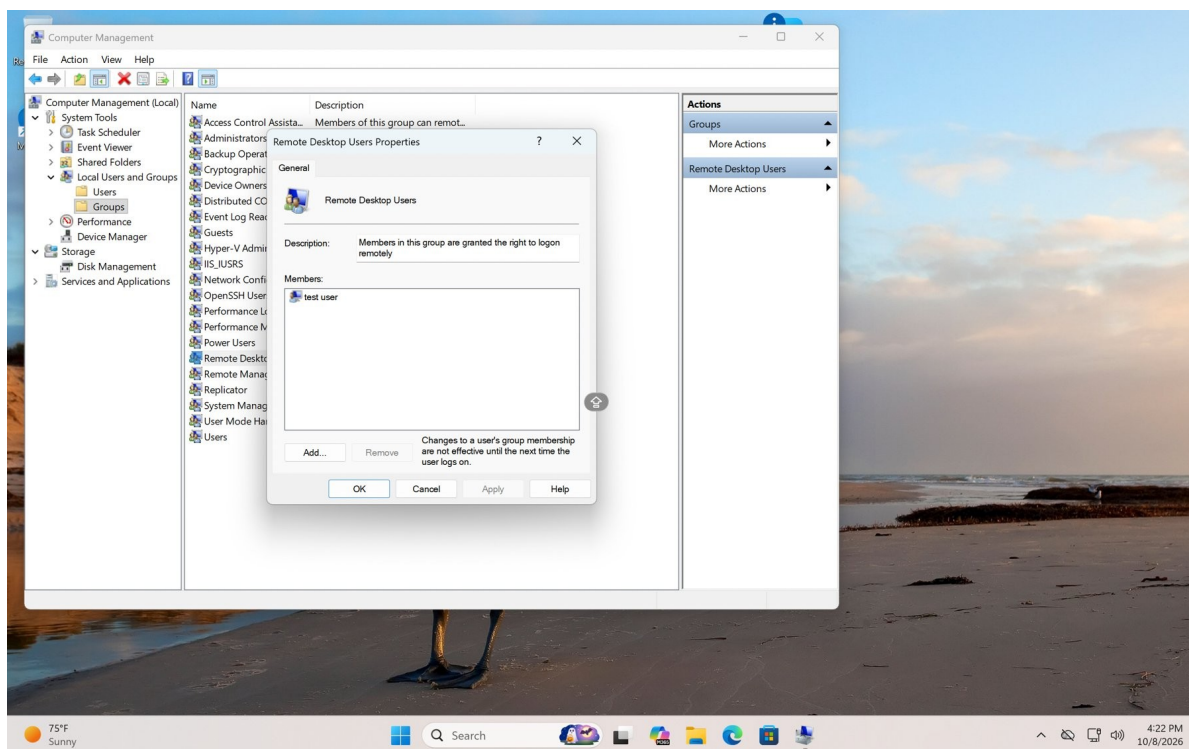
Lab Environment

- Host computer: Mac running VMware Fusion
- Virtual machine operating system: Windows 11 Pro (version reported during setup)
- Lab scope: Local Windows administration and troubleshooting
- Note: This was a Windows 11 lab, not a Windows Server deployment or an Active Directory domain-controller build.

1. Local Users and Groups

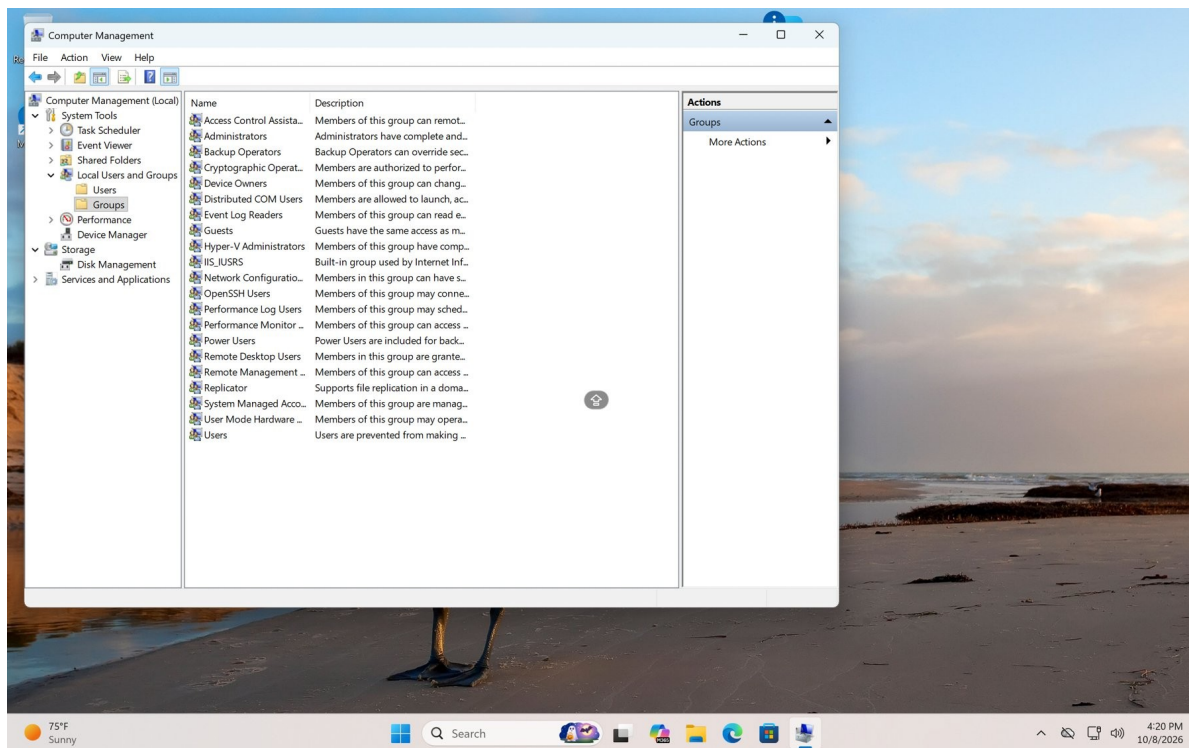
Used Computer Management → Local Users and Groups to inspect local group membership. The test account was shown as a member of the Remote Desktop Users group. This demonstrates familiarity with Windows local account and group-management concepts.

Evidence 1 — Remote Desktop Users Membership



Remote Desktop Users Properties showing "test user" as a member

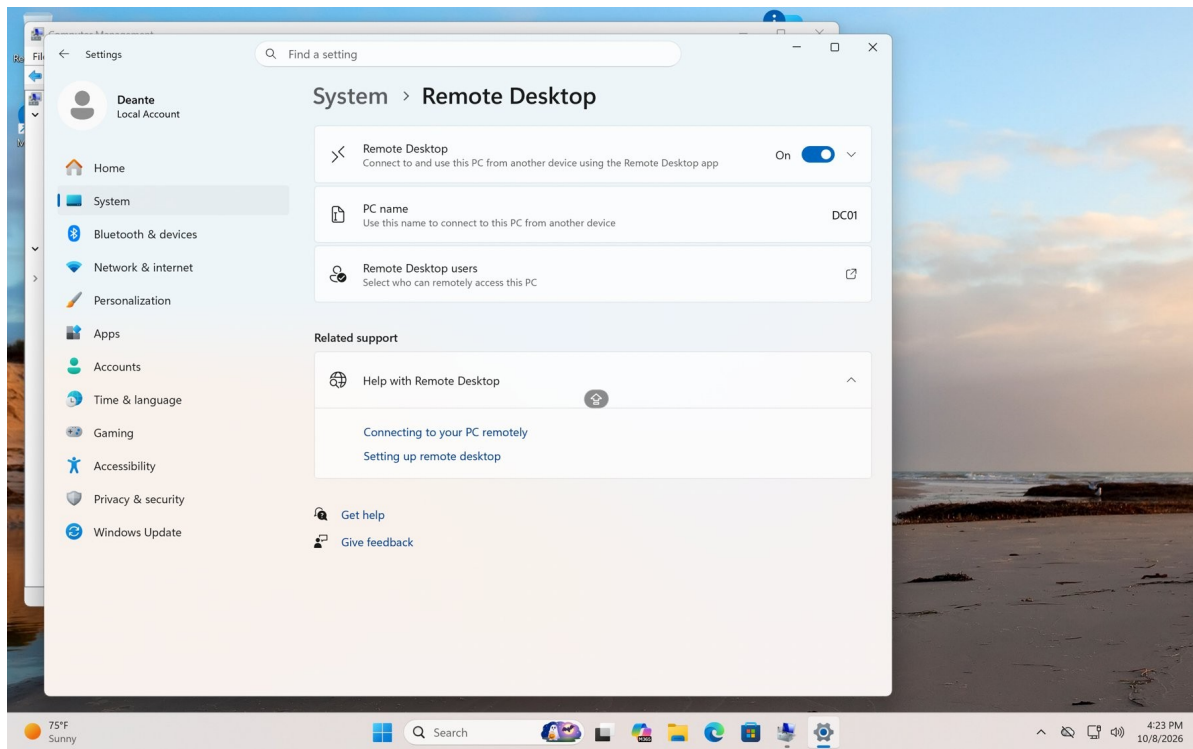
Evidence 2 — Local Groups Overview



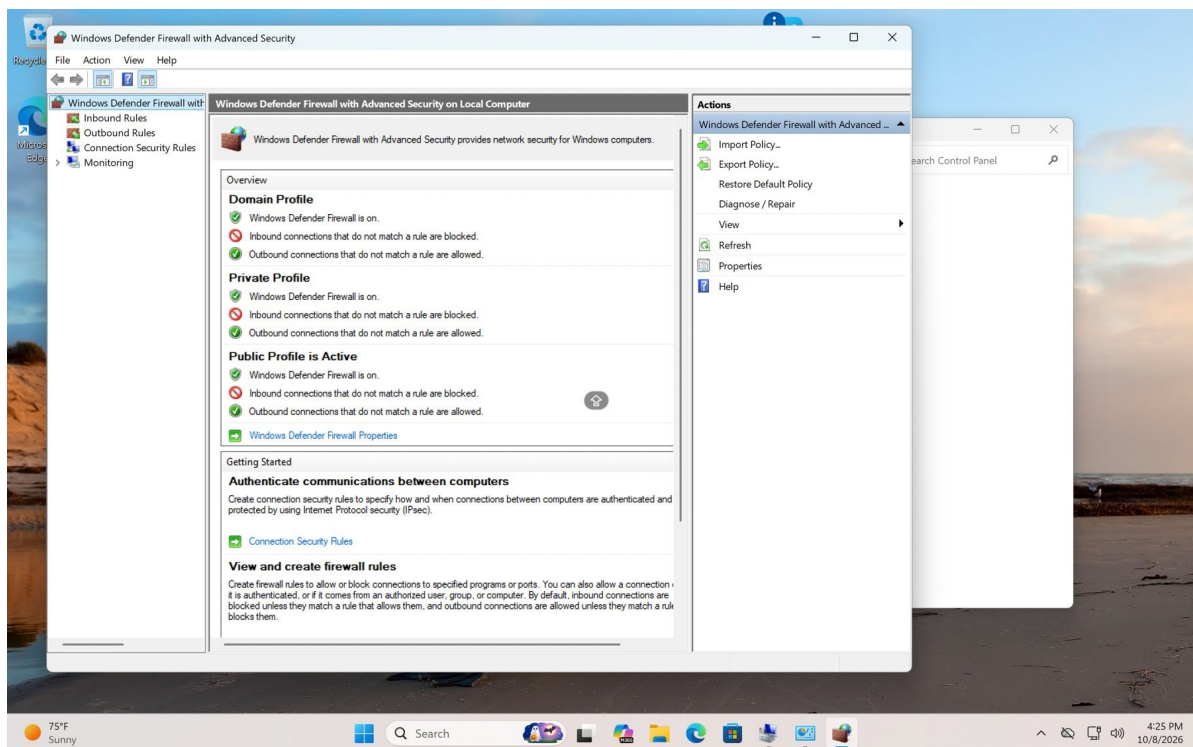
Computer Management showing the local Groups list

2. Remote Desktop and Firewall Review

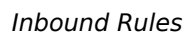
Reviewed Remote Desktop configuration and inspected Windows Defender Firewall with Advanced Security, including inbound and outbound rules. The firewall rules were reviewed for familiarization; no changes should be claimed unless they were actually made.



Settings > System > Remote Desktop (Remote Desktop turned On)

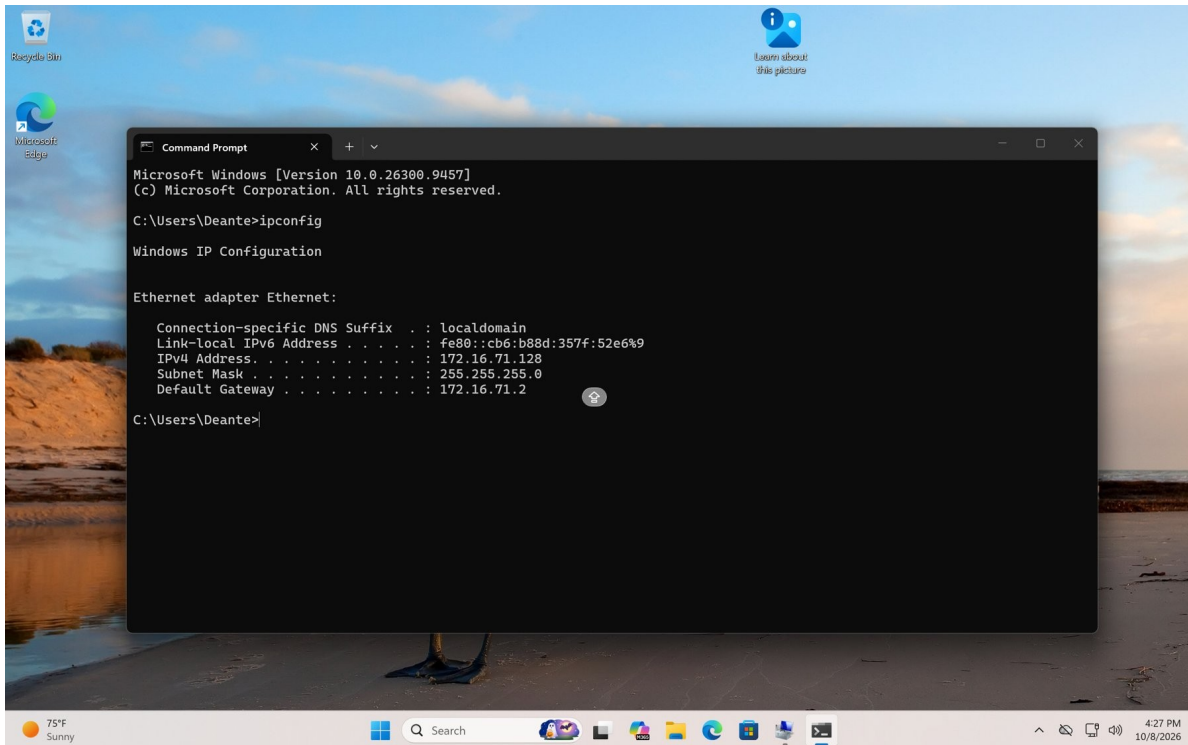


Windows Defender Firewall with Advanced Security overview



3. Network Troubleshooting

Used ipconfig and ipconfig /all to inspect network configuration. Used ping 8.8.8.8 to test IP connectivity and ping google.com to test name resolution and connectivity. Record the actual displayed results in the final version if they are available.



A screenshot of a Windows desktop with a beach background. A Command Prompt window is open, displaying the output of the 'ipconfig' command. The desktop has icons for Recycle Bin, Microsoft Edge, and a picture. The taskbar shows the Start button, Search, and several application icons. The system tray indicates 75°F Sunny and the date 10/8/2026.

```
Microsoft Windows [Version 10.0.26300.9457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Deante>ipconfig

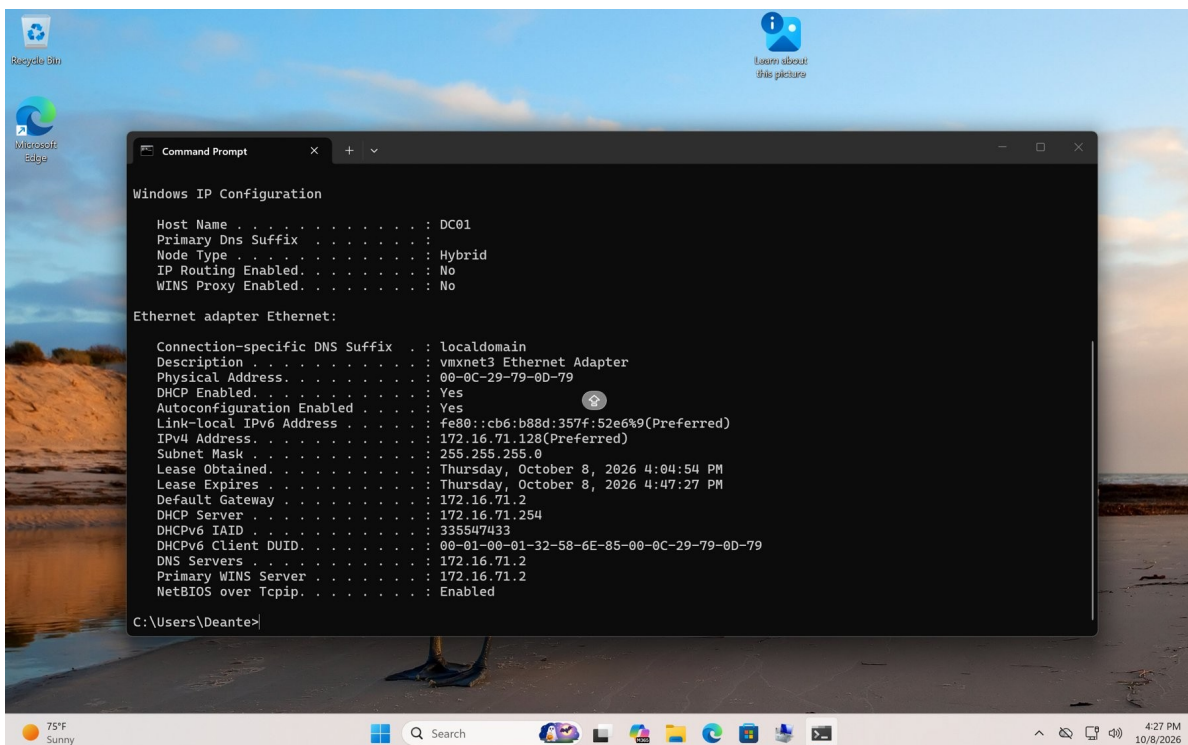
Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : localdomain
    Link-local IPv6 Address . . . . . : fe80::cb6:b88d:357f:52e6%9
    IPv4 Address. . . . . : 172.16.71.128
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 172.16.71.2

C:\Users\Deante>
```

ipconfig output



A screenshot of a Windows desktop with a beach background. A Command Prompt window is open, displaying the output of the 'ipconfig /all' command. The desktop has icons for Recycle Bin, Microsoft Edge, and a picture. The taskbar shows the Start button, Search, and several application icons. The system tray indicates 75°F Sunny and the date 10/8/2026.

```
Windows IP Configuration

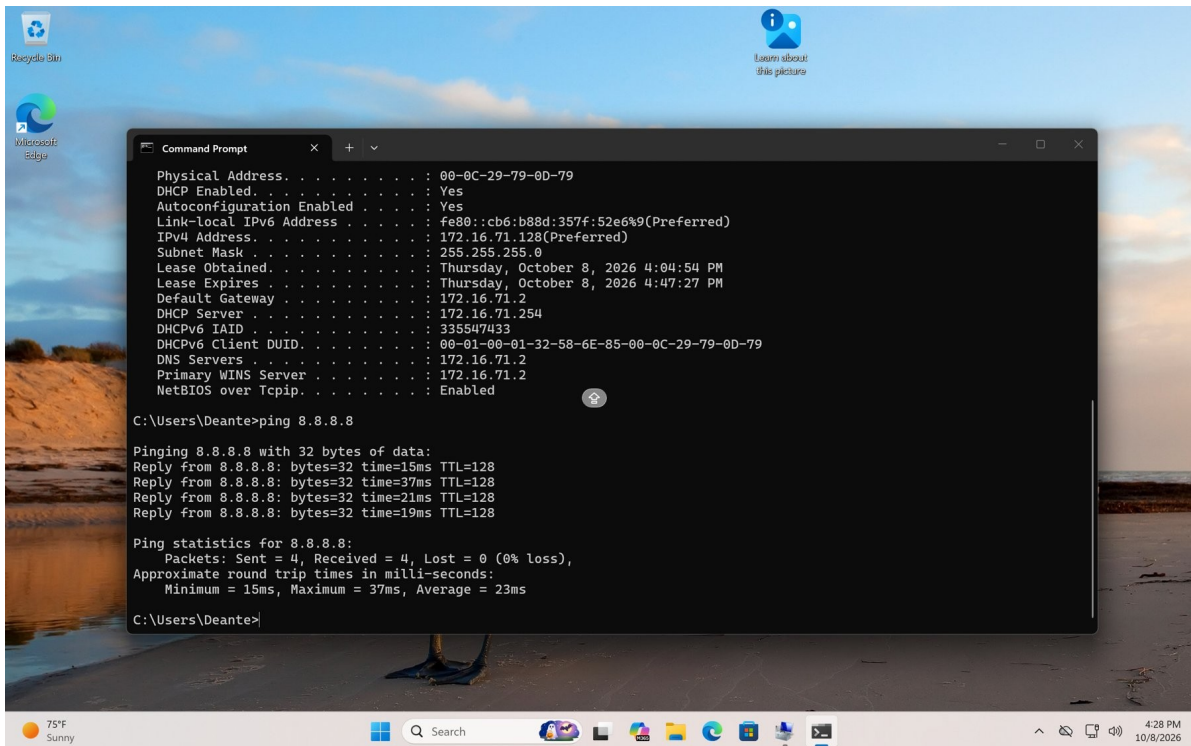
Host Name . . . . . : DC01
Primary Dns Suffix . . . . . : 
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

Ethernet adapter Ethernet:

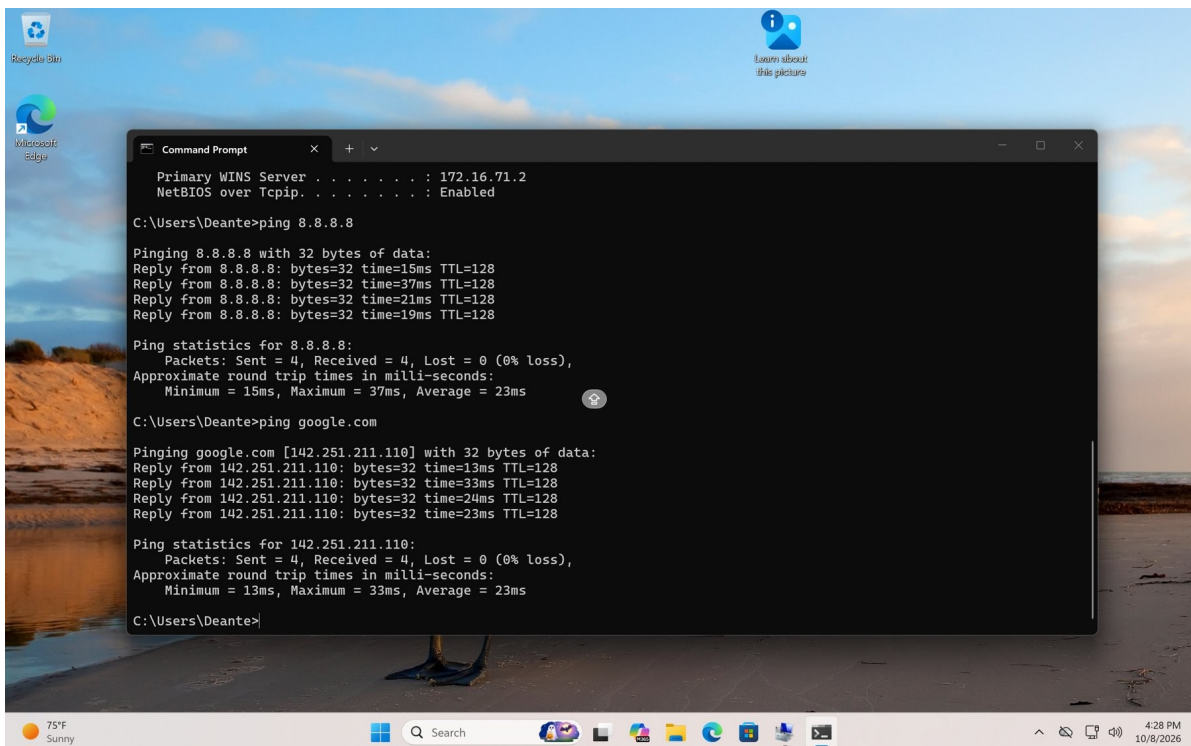
    Connection-specific DNS Suffix  . : localdomain
    Description . . . . . : vmxnet3 Ethernet Adapter
    Physical Address. . . . . : 00-0C-29-79-0D-79
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes
    Link-local IPv6 Address . . . . . : fe80::cb6:b88d:357f:52e6%9(Preferred)
    IPv4 Address. . . . . : 172.16.71.128(Preferred)
    Subnet Mask . . . . . : 255.255.255.0
    Lease Obtained. . . . . : Thursday, October 8, 2026 4:04:54 PM
    Lease Expires . . . . . : Thursday, October 8, 2026 4:47:27 PM
    Default Gateway . . . . . : 172.16.71.2
    DHCP Server . . . . . : 172.16.71.254
    DHCPv6 IAID . . . . . : 335547433
    DHCPv6 Client DUID. . . . . : 00-01-00-01-32-58-6E-85-00-0C-29-79-0D-79
    DNS Servers . . . . . : 172.16.71.2
    Primary WINS Server . . . . . : 172.16.71.2
    NetBIOS over Tcpip. . . . . : Enabled

C:\Users\Deante>
```

ipconfig /all output



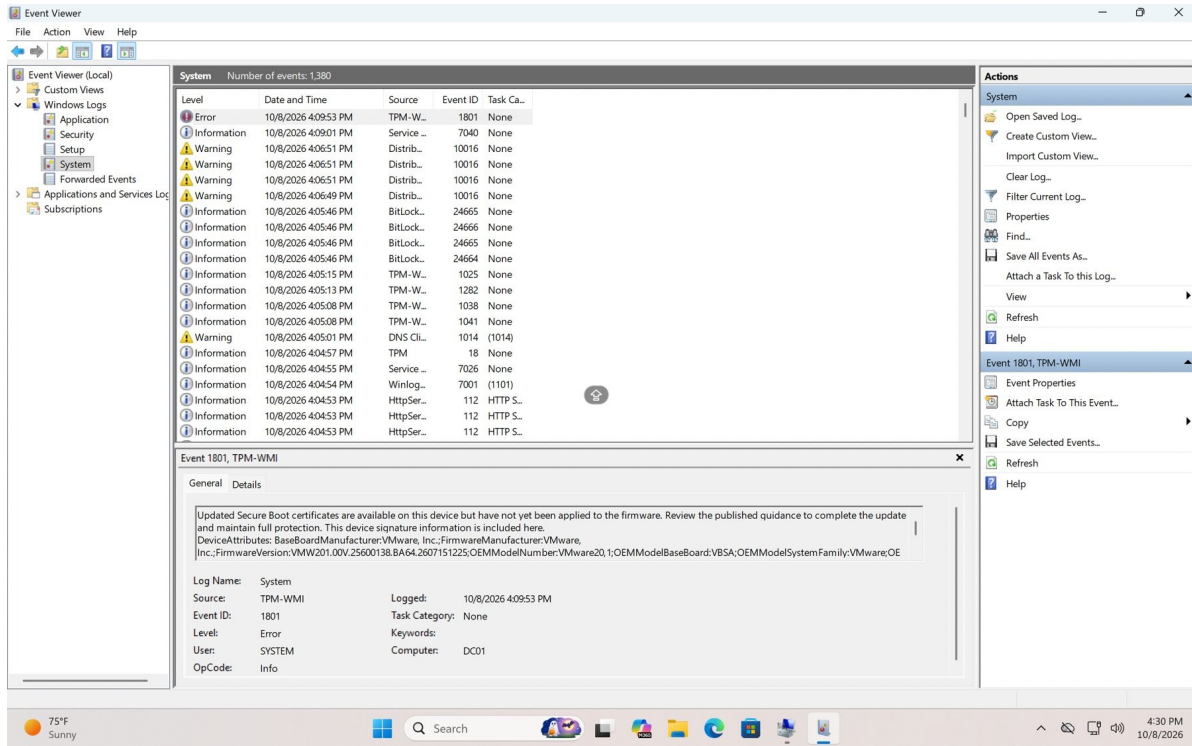
ping 8.8.8.8 (0% packet loss)



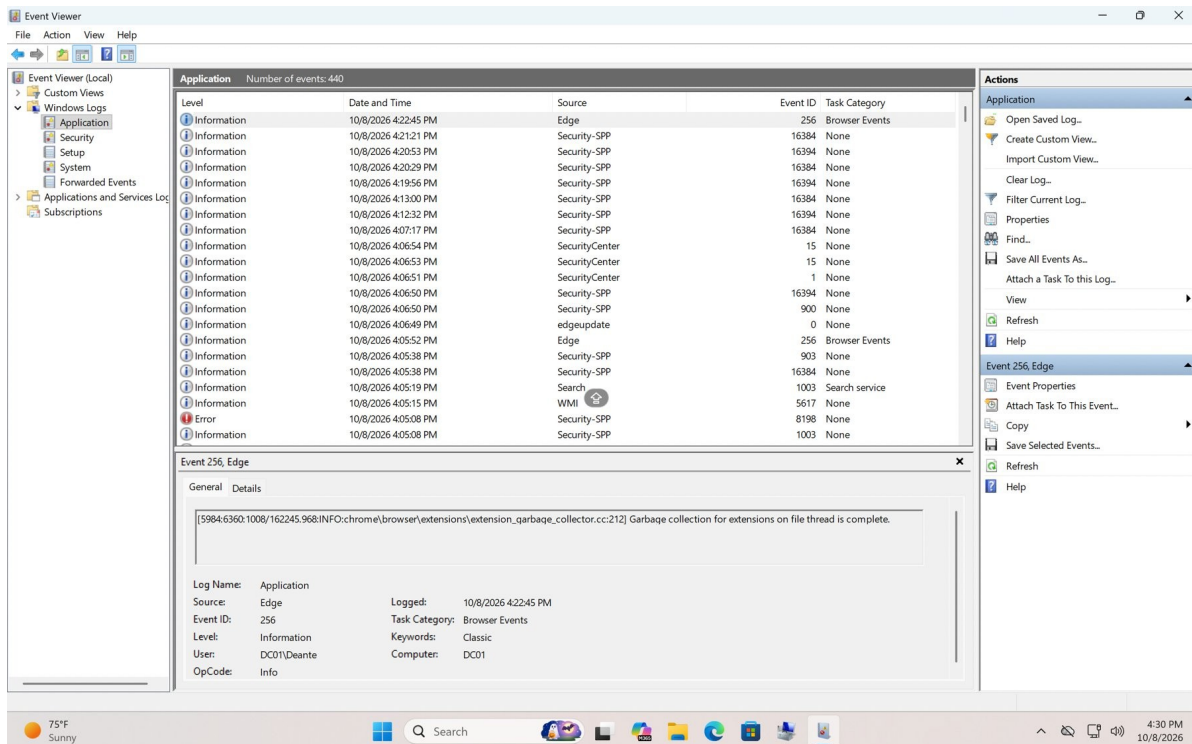
ping google.com (name resolved, 0% packet loss)

4. Event Viewer Review

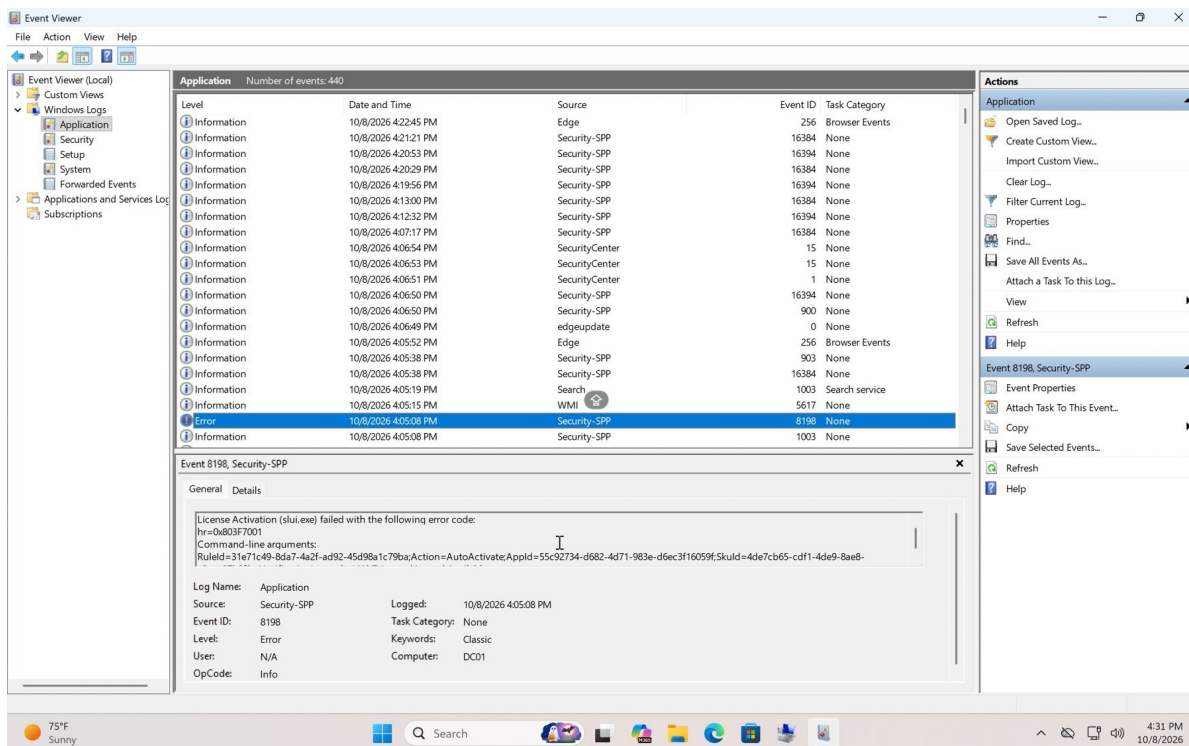
Reviewed Windows Logs → System and Application in Event Viewer. An Application log entry was inspected with Source Security-SPP and Event ID 8198. This entry can relate to software licensing/activation; it was reviewed as a troubleshooting exercise rather than treated as proof of a broader system failure.



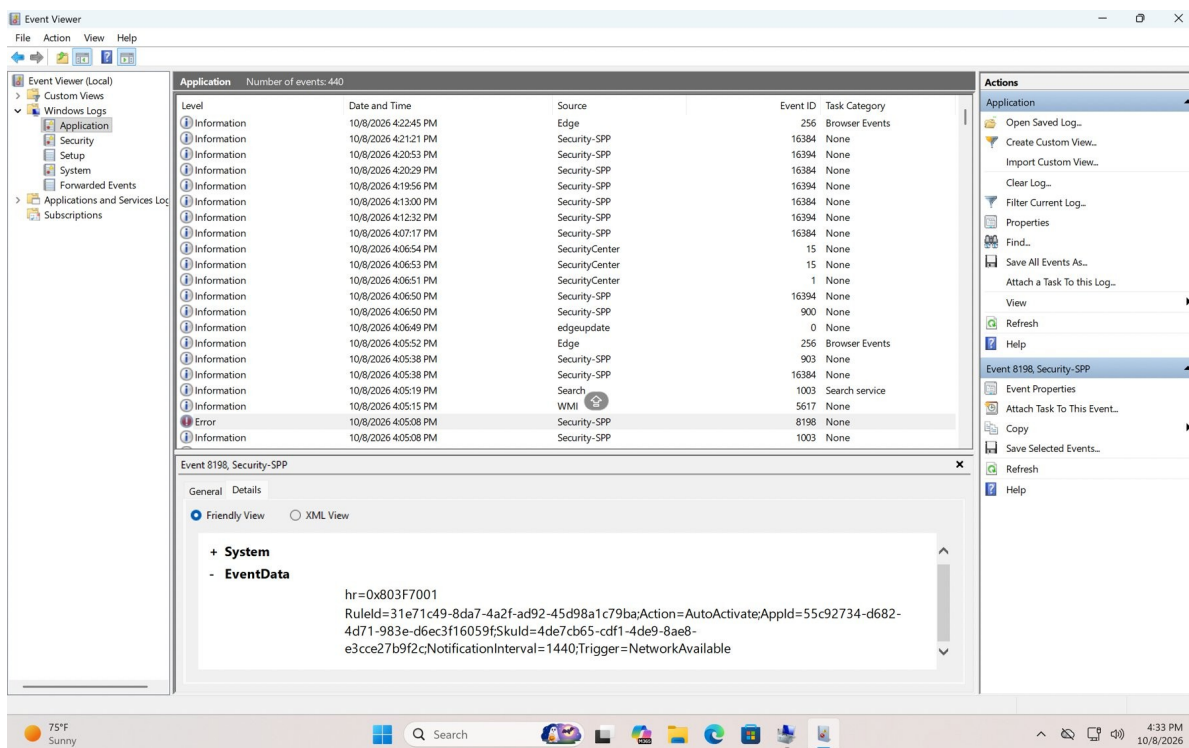
Event Viewer: Windows Logs > System



Event Viewer: Windows Logs > Application



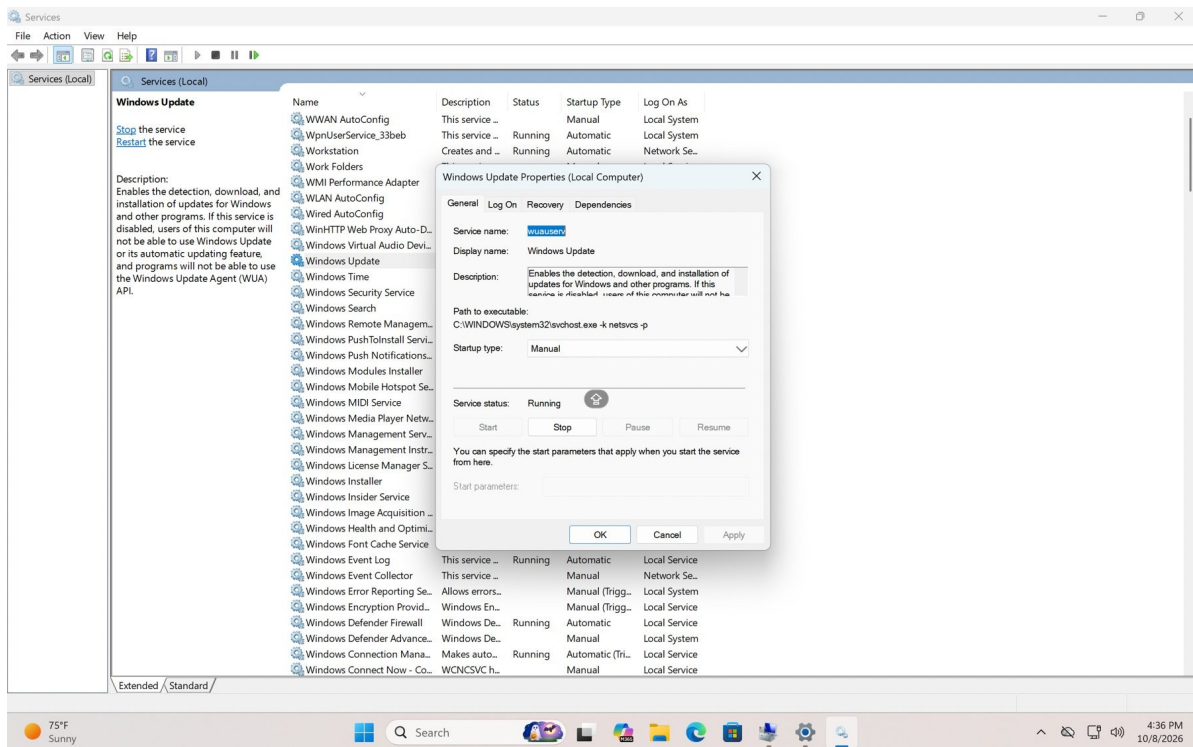
Application log event 8198, Security-SPP: General tab



Application log event 8198, Security-SPP: Details tab

5. Services, Performance, and Storage

Opened Services and reviewed Windows Update Properties; the service status was reported as Running. Reviewed Task Manager → Performance, Disk Management, and System Information. Disk Management was inspected without changing partitions.



Windows Update Properties (service status: Running)



Task Manager > Performance

Disk Management

File Action View Help

Volume	Layout	Type	File System	Status	Capacity	Free Sp...	% Free
(C:)	Simple	Basic	NTFS (BitLo...	Healthy (B...	61.29 GB	35.83 GB	58 %
(Disk 0 partition 1)	Simple	Basic		Healthy (E...	200 MB	200 MB	100 %
(Disk 0 partition 4)	Simple	Basic		Healthy (R...	853 MB	853 MB	100 %
(Disk 0 partition 5)	Simple	Basic		Healthy (R...	853 MB	853 MB	100 %
(Disk 0 partition 6)	Simple	Basic		Healthy (R...	853 MB	853 MB	100 %
SSS_X64FREE_EN-...	Simple	Basic	UDF	Healthy (P...	7.59 GB	0 MB	0 %

Disk 0
Basic
63.98 GB
Online

200 MB
Healthy (EFI System Partition)

(C:)
61.29 GB NTFS (BitLocker Encrypted)
Healthy (Boot, Page File, Crash Dump, Basic Data Partition)

853 MB
Healthy (Recovery Partition)

853 MB
Healthy (Recovery Partition)

853 MB
Healthy (Recovery Partition)

CD-ROM 0
DVD
7.59 GB
Online

SSS_X64FREE_EN-US_DV9 (D:)
7.59 GB UDF
Healthy (Primary Partition)

■ Unallocated ■ Primary partition

75°F Sunny

Search

4:39 PM 10/8/2026

Disk Management

System Information

File Edit View Help

System Summary

- Hardware Resources
- Components
- Software Environment

Item	Value
OS Name	Microsoft Windows 11 Pro
Version	10.0.26300 Build 26300
Other OS Description	Not Available
OS Manufacturer	Microsoft Corporation
System Name	DC01
System Manufacturer	VMware, Inc.
System Model	VMware20.1
System Type	ARM64-based PC
System SKU	0000000000000001
Processor	Apple silicon, 2000 Mhz, 1 Core(s), 1 Logical Processor(s)
Processor	Apple silicon, 2000 Mhz, 1 Core(s), 1 Logical Processor(s)
BIOS Version/Date	VMware, Inc. VMW201.00V.25600138.BA64.2607151225, 7/15/2026
SMBIOS Version	3.2
Embedded Controller Version	255.255
BIOS Mode	UEFI
BaseBoard Manufacturer	VMware, Inc.
BaseBoard Product	VB5A
BaseBoard Version	1
Platform Role	Desktop
Secure Boot State	On
PCR7 Configuration	Elevation Required to View
Windows Directory	C:\WINDOWS
System Directory	C:\WINDOWS\system32
Boot Device	\Device\HarddiskVolume1
Locale	United States
Hardware Abstraction Layer	Version = "10.0.26100.1"
User Name	DC01\Deante
Time Zone	Pacific Daylight Time
Installed Physical Memory (RAM)	4.00 GB
Total Physical Memory	3.99 GB
Available Physical Memory	1.28 GB
Total Virtual Memory	5.37 GB
Available Virtual Memory	2.52 GB
Page File Space	1.38 GB
Page File	C:\pagefile.sys
Kernel DMA Protection	Off

Find

Search selected category Search category names

Find Close Find

75°F Sunny

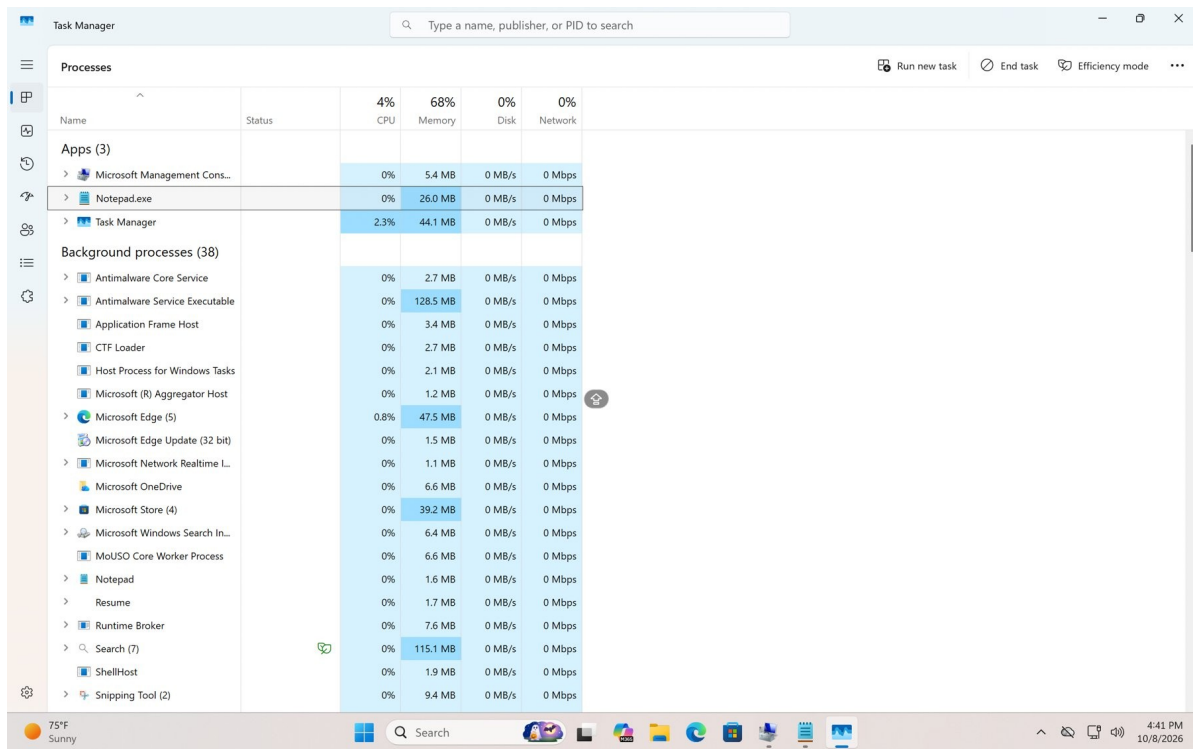
Search

4:39 PM 10/8/2026

System Information

6. Process Troubleshooting

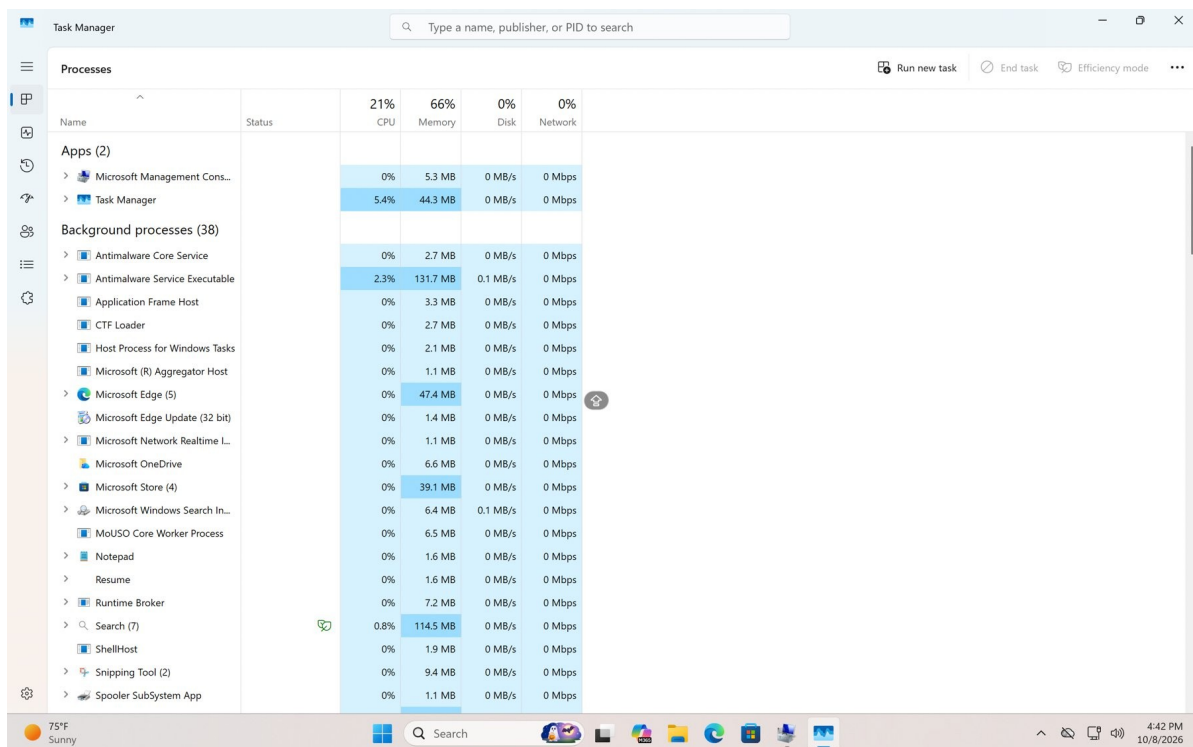
Opened Notepad and located Notepad.exe in Task Manager, then used End task to practice identifying and ending a process. This is a basic example of process monitoring and application troubleshooting.



The screenshot shows the Windows Task Manager window with the 'Processes' tab selected. The 'Apps' section is expanded, showing a list of running applications. Notepad.exe is listed with 0% CPU usage and 26.0 MB of memory. The 'Background processes' section is also expanded, showing various system services and background applications.

Name	Status	4% CPU	68% Memory	0% Disk	0% Network
Apps (3)					
Microsoft Management Cons...		0%	5.4 MB	0 MB/s	0 Mbps
Notepad.exe		0%	26.0 MB	0 MB/s	0 Mbps
Task Manager		2.3%	44.1 MB	0 MB/s	0 Mbps
Background processes (38)					
Antimalware Core Service		0%	2.7 MB	0 MB/s	0 Mbps
Antimalware Service Executable		0%	128.5 MB	0 MB/s	0 Mbps
Application Frame Host		0%	3.4 MB	0 MB/s	0 Mbps
CTF Loader		0%	2.7 MB	0 MB/s	0 Mbps
Host Process for Windows Tasks		0%	2.1 MB	0 MB/s	0 Mbps
Microsoft (R) Aggregator Host		0%	1.2 MB	0 MB/s	0 Mbps
Microsoft Edge (5)		0.8%	47.5 MB	0 MB/s	0 Mbps
Microsoft Edge Update (32 bit)		0%	1.5 MB	0 MB/s	0 Mbps
Microsoft Network Realtime L...		0%	1.1 MB	0 MB/s	0 Mbps
Microsoft OneDrive		0%	6.6 MB	0 MB/s	0 Mbps
Microsoft Store (4)		0%	39.2 MB	0 MB/s	0 Mbps
Microsoft Windows Search In...		0%	6.4 MB	0 MB/s	0 Mbps
MoUSO Core Worker Process		0%	6.6 MB	0 MB/s	0 Mbps
Notepad		0%	1.6 MB	0 MB/s	0 Mbps
Resume		0%	1.7 MB	0 MB/s	0 Mbps
Runtime Broker		0%	7.6 MB	0 MB/s	0 Mbps
Search (7)		0%	115.1 MB	0 MB/s	0 Mbps
ShellHost		0%	1.9 MB	0 MB/s	0 Mbps
Snipping Tool (2)		0%	9.4 MB	0 MB/s	0 Mbps

Notepad.exe listed under Apps in Task Manager



The screenshot shows the Windows Task Manager window after Notepad.exe has been ended. The 'Apps' section now only contains Microsoft Management Console and Task Manager. The 'Background processes' section remains expanded, showing the same list of system services and background applications as before.

Name	Status	21% CPU	66% Memory	0% Disk	0% Network
Apps (2)					
Microsoft Management Cons...		0%	5.3 MB	0 MB/s	0 Mbps
Task Manager		5.4%	44.3 MB	0 MB/s	0 Mbps
Background processes (38)					
Antimalware Core Service		0%	2.7 MB	0 MB/s	0 Mbps
Antimalware Service Executable		2.3%	131.7 MB	0.1 MB/s	0 Mbps
Application Frame Host		0%	3.3 MB	0 MB/s	0 Mbps
CTF Loader		0%	2.7 MB	0 MB/s	0 Mbps
Host Process for Windows Tasks		0%	2.1 MB	0 MB/s	0 Mbps
Microsoft (R) Aggregator Host		0%	1.1 MB	0 MB/s	0 Mbps
Microsoft Edge (5)		0%	47.4 MB	0 MB/s	0 Mbps
Microsoft Edge Update (32 bit)		0%	1.4 MB	0 MB/s	0 Mbps
Microsoft Network Realtime L...		0%	1.1 MB	0 MB/s	0 Mbps
Microsoft OneDrive		0%	6.6 MB	0 MB/s	0 Mbps
Microsoft Store (4)		0%	39.1 MB	0 MB/s	0 Mbps
Microsoft Windows Search In...		0%	6.4 MB	0.1 MB/s	0 Mbps
MoUSO Core Worker Process		0%	6.5 MB	0 MB/s	0 Mbps
Notepad		0%	1.6 MB	0 MB/s	0 Mbps
Resume		0%	1.6 MB	0 MB/s	0 Mbps
Runtime Broker		0%	7.2 MB	0 MB/s	0 Mbps
Search (7)		0.8%	114.5 MB	0 MB/s	0 Mbps
ShellHost		0%	1.9 MB	0 MB/s	0 Mbps
Snipping Tool (2)		0%	9.4 MB	0 MB/s	0 Mbps
Spooler SubSystem App		0%	1.1 MB	0 MB/s	0 Mbps

Task Manager after using End task on Notepad.exe

Skills Practiced

- Windows Computer Management and local group membership
- Remote Desktop configuration concepts
- Windows Defender Firewall rule inspection
- Basic network diagnostics using ipconfig and ping
- Event Viewer log review
- Windows Services inspection
- Task Manager process and performance monitoring
- Disk Management and System Information review
- Basic application-process troubleshooting

Portfolio / Resume Description

Completed a hands-on Windows 11 administration and troubleshooting lab in VMware Fusion. Practiced local user/group management, reviewed firewall rules and Windows event logs, ran basic network connectivity tests, inspected Windows services and system performance, and used Task Manager to investigate an application process.

This report documents a personal practice lab in a virtual machine and is not professional employment experience.